

A Multimodal Stroke Prediction System Using Smartphone Photoplethysmography, Self-Reported Risk Factors, and Medical Document Analysis

Abstract—Absolute stroke risk estimation is conventionally confined to clinical settings because it requires measurements that individuals cannot obtain unaided. We present a multimodal system that produces a calibrated five-year absolute stroke probability from three inputs available to any smartphone user: a structured questionnaire, a sixty-second fingertip photoplethysmogram acquired with the device camera and torch, and optional photographs of medication packaging or laboratory reports. Absolute risk is anchored to a published Cox proportional-hazards model derived from a prospective cohort of 23,691 adults with adjudicated stroke outcomes, extended with covariate-adjusted adjunct terms for population-specific risk factors. Unanswered questions are propagated as distributions rather than imputed, yielding an interval estimate together with a variance-based ranking of which unmeasured input would most reduce uncertainty. Rhythm assessment uses a classical interval-statistics classifier trained on the MIT-BIH Atrial Fibrillation Database, achieving an area under the receiver operating characteristic curve of 0.990 under leave-one-patient-out validation. We introduce an arrhythmia-aware signal quality assessment, motivated by the observation that the spectral and rate-stability metrics conventionally used to gate photoplethysmographic heart-rate estimation are structurally incompatible with arrhythmia detection. Independent evaluation on 300 smartphone recordings with synchronous electrocardiographic ground truth yields 90% beat-detection sensitivity, 90% positive predictive value, and 28.4 ms inter-beat interval error on expert-graded recordings. Evaluated at the deployed sixty-second window on fingertip photoplethysmography from 35 adults with rhythm labels, the classifier attains 97.0% sensitivity and 95.6% specificity, statistically unchanged when the signal is decimated to the 30 Hz frame rate of a smartphone camera. Screening performance gives a negative predictive value above 99.9% at population prevalence, supporting an asymmetric decision rule in which a regular rhythm resolves atrial fibrillation within the risk model while an irregular rhythm produces a referral rather than a classification.

Index Terms—stroke risk prediction, photoplethysmography, atrial fibrillation screening, mobile health, survival analysis, uncertainty quantification, optical character recognition

I. INTRODUCTION

Stroke is among the leading causes of death and acquired disability worldwide, and a substantial fraction of its burden is attributable to modifiable risk factors. Established risk equations such as the Framingham Stroke Risk Profile translate a set of clinical measurements into an absolute probability of incident stroke over a defined horizon. Their practical reach is limited by the inputs they require: systolic blood pressure measured by a trained observer, laboratory-confirmed diabetes status, and electrocardiographically documented atrial fibrillation.

Two consequences follow. First, individuals outside regular clinical contact cannot obtain an estimate at all. Second, where such tools are made available as self-assessment instruments, unknown inputs are typically replaced by population averages, and the resulting number is presented with a confidence that its inputs do not support.

This work addresses both. We describe a system that produces a calibrated five-year absolute stroke probability entirely on a consumer smartphone, combining three modalities that require no hardware beyond the device itself, and that represents what the user does not know as uncertainty rather than as an assumption.

The principal contributions are:

- 1) An *equation-anchored fusion architecture* in which absolute calibration derives from a published prospective-cohort survival model, and each additional modality either resolves a variable the model already contains or contributes a documented, covariate-adjusted adjunct term. Every coefficient remains traceable to a published source.
- 2) *Uncertainty propagation with value-of-information ranking*. Unanswered questions are carried as prior distributions and propagated by Monte Carlo, producing an interval rather than a point estimate. First-order Sobol

indices rank the unmeasured inputs by the share of output variance each explains, converting missing data into an actionable measurement recommendation.

- 3) *Arrhythmia-aware signal quality assessment*. We show that spectral signal-to-noise ratio and inter-window rate consistency—the two metrics conventionally used to gate photoplethysmographic heart-rate estimation—are structurally unsuitable for gating arrhythmia analysis, and give a replacement based on band-aggregate and time-domain criteria.
- 4) An *asymmetric decision rule* derived from the predictive-value asymmetry of low-prevalence screening, under which the rhythm module may exclude atrial fibrillation from the risk computation but never assert it.
- 5) *Population-specific extension*, demonstrated by incorporating smokeless tobacco use—a dominant exposure in South Asia that Western-derived equations omit—as a separately weighted adjunct term.

II. RELATED WORK

A. Absolute Risk Equations

The Framingham Stroke Risk Profile [1] established points-based estimation of ten-year stroke probability from age, systolic blood pressure, antihypertensive treatment, diabetes, smoking, prevalent cardiovascular disease, atrial fibrillation and electrocardiographic left ventricular hypertrophy. A revision [2] recalibrated the model against contemporary cohorts, reporting concordance statistics between 0.62 and 0.78 across derivation and external validation samples. Composite cardiovascular instruments such as the Pooled Cohort Equations and QRISK3 include stroke within a broader endpoint.

B. Machine Learning on Survey Data

Gradient-boosted and ensemble classifiers trained on large cross-sectional health surveys report discrimination exceeding 0.82 for stroke history. Such models predict *prevalent* self-reported stroke rather than incident events; because the training data carry no time axis, their output is not a forward risk and is not directly comparable to survival-model concordance.

C. Smartphone Photoplethysmography

Contact photoplethysmography using a fingertip placed over a smartphone camera with the torch illuminated has been validated for atrial fibrillation screening in several prospective studies, with reported sensitivity between 91.5% and 96.3% and specificity between 97% and 99.6% [3], [4]. A meta-analysis of ten diagnostic accuracy studies covering 3,852 participants found that, when screening asymptomatic populations, positive predictive value falls to approximately 20–40% while negative predictive value approaches 100% [5]. This asymmetry is a direct consequence of low disease prevalence and motivates the decision rule adopted here.

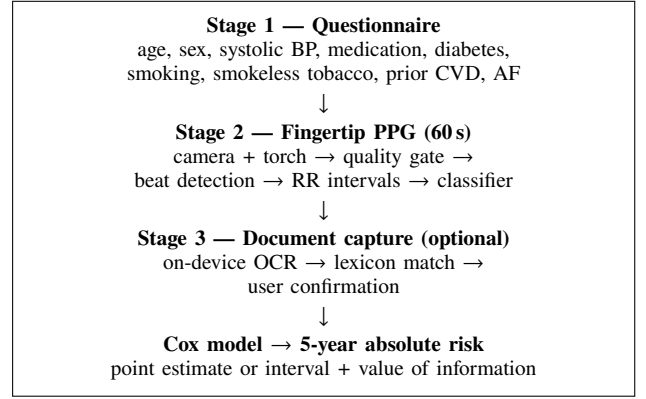


Fig. 1. Three-stage acquisition. Each stage may only resolve a variable left unknown by the preceding stage; no stage overrides a direct user response.

D. Interval-Based Arrhythmia Detection

Detection of atrial fibrillation from inter-beat intervals is well established using statistical irregularity descriptors—root mean square of successive differences, Shannon entropy, sample entropy [6] and Poincaré descriptors. These operate on interval sequences and are therefore agnostic to whether those intervals originate from electrocardiography or photoplethysmography, which permits the classifier and the acquisition path to be validated separately.

III. SYSTEM ARCHITECTURE

The system executes entirely within the browser on the user's device. No questionnaire response, video frame or photograph is transmitted. Fig. 1 summarises the flow.

A single invariant governs fusion: *a later stage may resolve an unknown, but never overrides an explicit user response, and never asserts a positive finding that its evidence cannot support*.

IV. ABSOLUTE RISK ESTIMATION

A. Model Form

Five-year risk is computed from a Cox proportional-hazards model,

$$\hat{p}(t = 5 | \mathbf{x}) = 1 - S_0(5)^{\exp(\text{LP}(\mathbf{x}) - M)}, \quad (1)$$

where $\text{LP}(\mathbf{x}) = \sum_j \beta_j x_j$, $\beta_j = \ln(\text{HR}_j)$, and $M = \sum_j \beta_j \bar{x}_j$ centres the predictor on the derivation cohort mean. Diabetes enters through one of two age-conditioned coefficients, reflecting the single retained interaction in the source model; systolic pressure enters through a treated or untreated coefficient according to medication status.

B. Coefficient Selection

Two coefficient sets are published for the revised profile. We adopt the REGARDS estimates as the operational default on four grounds: the derivation sample is 23,691 against 5,072; follow-up is exactly five years, matching the prediction horizon without extrapolation; events were adjudicated under six-monthly active surveillance; and the cohort includes substantial representation permitting race-stratified baseline rates. Table I lists the hazard ratios.

TABLE I
HAZARD RATIOS, REGARDS DERIVATION SAMPLE, AGE ≥ 55

Risk factor	Men	Women
Age (per 10 years)	1.72	1.83
Current smoking	2.00	1.77
Prevalent CVD	1.31	1.52
Atrial fibrillation	1.62	1.57
Age ≥ 65	1.34	1.44
Diabetes, age < 65	1.94	1.66
Diabetes, age ≥ 65	1.15	1.11
Antihypertensive treatment	1.26	1.19
SBP, untreated (per 10 mmHg)	1.09	1.16
SBP, treated (per 10 mmHg)	1.05	1.18
<i>Adjunct term</i>		
Smokeless tobacco	1.35	1.35

C. Recovery of Baseline Survival

The source publication reports Kaplan–Meier cohort risk but not S_0 . Because risk is convex in the linear predictor, the mean of risk exceeds risk at the mean, so equating S_0 with $1 - \text{KM}$ introduces a systematic bias. We instead construct a synthetic cohort matching the published covariate marginals and solve numerically for the S_0 reproducing the observed mean risk:

$$S_0(5) : \mathbb{E}_{\mathbf{x}}[1 - S_0(5)^{\exp(\text{LP}(\mathbf{x}) - M)}] = \text{KM}(5). \quad (2)$$

This yields $S_0(5) = 0.97310$ for men and 0.97803 for women. The direct substitution would give 0.969 and 0.973 , overstating risk for a cohort-representative individual by approximately 15% in relative terms. Reconstruction accuracy is verified by confirming that the mean predicted risk over an independently seeded synthetic cohort reproduces the published five-year Kaplan–Meier estimates of 0.031 and 0.027 to within 0.002 .

D. Adjunct Terms

Risk factors absent from the derivation cohort are incorporated as additional linear-predictor terms using published covariate-adjusted hazard ratios. Such terms are centred at zero rather than at the target-population prevalence: the derivation cohort is the reference against which S_0 was calibrated, and where an exposure was negligible in that cohort, non-exposure already constitutes the reference level. Centring on target prevalence would apply an unwarranted reduction to unexposed individuals. Cohort calibration is verified to be unaffected.

We apply this to smokeless tobacco, with a pooled stroke relative risk of 1.35 (95% CI 1.20 – 1.50) for chewing-tobacco users [7]. The same meta-analysis reports the overall association to be essentially unchanged under strict adjustment for smoking (1.17 unadjusted versus 1.18 adjusted), supporting treatment of this exposure as separable from combustible tobacco. It is queried independently because its effect is materially smaller than that of smoking (HR 2.00 in men); combining the two would overstate risk for exclusive smokeless users by approximately 50%.

E. Uncertainty Propagation

Where a response is unknown, the variable is sampled from a population prior and propagated by Monte Carlo, and the system reports the 10th–90th percentile interval. A fully specified profile collapses the interval to a point.

To convert missing data into guidance, we compute the first-order Sobol index for each unknown input,

$$S_j = \frac{\text{Var}(R) - \mathbb{E}_{x_j}[\text{Var}(R | X_j = x_j)]}{\text{Var}(R)}, \quad (3)$$

the share of output variance attributable to that input. Variance-based decomposition is bounded in $[0, 1]$ and is well defined for the non-additive response surface, and the resulting ranking is subject-specific rather than fixed.

F. Counterfactual Guidance

Modifiable factors are reported with the risk each change would remove. Changes in medication *status* are deliberately excluded from this set. At fixed measured systolic pressure the fitted model assigns treated individuals a lower hazard than untreated individuals, reflecting the flatter observed pressure–risk gradient in treated cohorts rather than a causal treatment effect; presenting it as an actionable lever would misrepresent the evidence. The pressure value itself is offered instead.

V. RHYTHM ASSESSMENT FROM SMARTPHONE PHOTOPLETHYSMOGRAPHY

A. Acquisition

The user places a fingertip over the rear camera and torch for sixty seconds. Illuminated contact photoplethysmography is used exclusively; ambient-light facial measurement is not offered, as it depends on skin reflectance and is markedly more sensitive to motion and illumination change. Per-frame mean intensity is computed over a central region of interest for each colour channel, and the channel carrying the cleanest pulse is selected automatically.

B. Sub-Sample Beat Localisation

At a typical capture rate of 30 frames s^{-1} , beat timing is quantised to 33 ms —comparable in magnitude to the beat-to-beat variability that distinguishes sinus rhythm from atrial fibrillation. Fitting a parabola through each candidate peak and its immediate neighbours recovers the extremum to a fraction of a frame. On a synthetic signal whose period is incommensurate with the frame interval, this reduces interval error from 15.9 ms to 0.04 ms , rendering frame rate a non-limiting factor.

C. Arrhythmia-Aware Quality Assessment

Photoplethysmographic quality gating conventionally relies on two families of metric: spectral signal-to-noise ratio, and consistency of the estimated rate across sliding windows. Both are appropriate when the objective is a heart rate. Both are structurally unsuitable when the objective is rhythm classification:

- **Inter-window rate consistency** penalises a rate that varies between windows. In atrial fibrillation the rate genuinely varies between windows; the criterion therefore rejects the target condition.
- **Spectral signal-to-noise ratio** quantifies the sharpness of the dominant spectral peak. Irregular intervals distribute pulse energy across the band, so atrial fibrillation has no sharp peak by construction. A clean synthetic fibrillatory waveform measures a negative signal-to-noise ratio under this definition.

Both metrics conflate the signal of interest with noise. We therefore gate rhythm analysis on criteria that are band-aggregate or purely temporal, and which respond to motion without responding to rhythm:

- 1) **In-band power fraction**: the proportion of detrended signal variance falling within the pulse band, which detects the presence of a pulse at *any* rate within the band rather than at one frequency.
- 2) **Photometric stability**: relative variation in mean frame intensity, which responds directly to finger displacement and illumination change.
- 3) **Beat amplitude consistency**: the coefficient of variation of detected peak amplitudes. Motion artefacts produce heterogeneous peak morphology; atrial fibrillation alters interval spacing while leaving pulse morphology comparatively uniform.
- 4) **Interval plausibility rate**: the proportion of detected intervals falling outside physiological bounds, distinguishing failed beat detection from genuine irregularity within plausible bounds.

D. Features and Classifier

Eight scale-free descriptors are computed over each analysis window: normalised root mean square of successive differences, coefficient of variation, the proportion of successive differences exceeding 50 ms and 70 ms, normalised Shannon entropy of the interval histogram, sample entropy, the Poincaré SD1/SD2 ratio, and mean rate. A standardised logistic regression is fitted over these features. A linear model is retained deliberately: the decision is auditable, the exported parameterisation is 4 kB, and inference requires no runtime dependency.

Sample entropy is undefined when no template pairs match within tolerance, a condition arising at both extremes of regularity—highly irregular sequences, where no matches exist, and near-constant sequences, where the matching tolerance approaches zero. We substitute the conventional upper bound $-\ln(2/[(N-m-1)(N-m)])$, mapping both limiting cases to maximal entropy and admitting the full range of rhythm into training.

E. Asymmetric Decision Rule

Because predictive value is prevalence-dependent, a screening instrument with high sensitivity and specificity may still yield a positive predictive value far below certainty. The module therefore emits one of three outcomes:

- **Regular**: atrial fibrillation is resolved to negative within the risk model, and the interval estimate narrows accordingly.
- **Irregular**: the risk estimate is left unchanged and the user is advised to seek electrocardiography. No classification is asserted.
- **Inconclusive**: quality criteria are not met; no inference is drawn.

This is not a conservatism heuristic but a direct consequence of the predictive values reported in Section VIII.

VI. MEDICAL DOCUMENT ANALYSIS

The optional third modality accepts a photograph of medication packaging, a prescription, or a laboratory report. Optical character recognition executes on-device, preserving the property that no image leaves the handset.

Recognised text is matched against a curated lexicon of 56 generic agents and 191 proprietary names, weighted toward South Asian markets where proprietary labelling predominates. Matching is tolerant to the substitution errors characteristic of curved, specular packaging surfaces, using normalised tokens compared under a length-dependent bounded edit distance. Laboratory values are extracted by proximity of a numeric literal to a recognised assay name, with range validation to reject spurious matches.

Each agent class maps to a model variable with an associated inference strength. Classes with a single dominant indication propose the corresponding variable directly. Classes with multiple indications—anticoagulants, which may be prescribed for fibrillation, venous thrombosis or prosthetic valves—raise a clarifying question rather than a value. Corroborating evidence from independent sources is accumulated into a single proposal.

No extracted value modifies the risk computation without explicit user confirmation. This constraint reduces recognition error from a correctness concern to an interaction concern: a misread becomes a question the user declines.

VII. EVALUATION METHODOLOGY

A. Rhythm Classifier

The classifier was trained and evaluated on the MIT-BIH Atrial Fibrillation Database, comprising 25 long-term recordings with verified beat annotations and rhythm episode labels [9]. Intervals were segmented into 60-second windows at 30-second hops; a window was labelled fibrillatory when at least 90% of its beats fell within an annotated episode, yielding 29,018 windows of which 37.2% were fibrillatory. Validation was leave-one-record-out, so no patient contributed to both training and evaluation.

B. Degradation Analysis

To characterise tolerance to acquisition error, features were recomputed from the annotated beat times after injecting timing perturbation and beat omission, and the full leave-one-record-out procedure was repeated at each level.

C. Acquisition Path

The acquisition path was evaluated independently on the Brno University of Technology Smartphone PPG Database, comprising fingertip recordings captured on consumer handsets at 30 frames s⁻¹ with synchronous 1 kHz electrocardiography and verified QRS annotations [8]. Analysis was restricted to fingertip acquisitions, yielding 300 recordings across eight annotated activity conditions.

Beat-detection accuracy was assessed by temporal correspondence rather than sequence subtraction: the pulse transit delay was estimated per recording, each detected peak matched to an electrocardiographic beat within a 100 ms tolerance, and interval error computed only across pairs whose endpoints both matched and were consecutive in both streams. This separates beat-detection sensitivity and positive predictive value from interval precision. The matching procedure was itself verified against synthetic sequences with known transit delay, known timing jitter, and deliberately omitted and inserted beats.

D. Deployed Window Length

The fingertip acquisition corpus provides ten-second clips, whereas the classifier operates over sixty seconds. Evaluation at the deployed configuration used an independent smartphone corpus recorded at the Gdańsk University of Technology [12], comprising one sixty-one-second acquisition with aligned electrocardiography and three continuous recordings of approximately ten minutes from healthy volunteers, segmented into non-overlapping sixty-second windows.

E. Rhythm Classification at the Deployed Configuration

The MIMIC PERform AF corpus [10] provides simultaneous fingertip photoplethysmography and electrocardiography from 35 critically ill adults, 19 in atrial fibrillation and 16 not, over twenty minutes each at 125 Hz. It is the only openly accessible rhythm-labelled corpus that is both fingertip—transmission through the digit, the optical geometry of a torch-illuminated camera capture—and long enough to segment at the deployed sixty-second window. Recordings were divided into non-overlapping sixty-second windows, discarding windows in which the sensor was disconnected, yielding 659 windows. To separate the contribution of frame rate from that of the sensing site, the evaluation was repeated on the same windows decimated to 30 Hz, the rate at which the system deploys. Rhythm labels are assigned per recording, so results are additionally aggregated per subject.

F. Rhythm Classification on Wrist Photoplethysmographic Intervals

Classifier sensitivity established on MIT-BIH is measured on intervals derived from the QRS complex. To measure it on intervals derived from a pulse wave instead, the frozen classifier was applied to the held-out test split of the DeepBeat corpus [11]: 17,617 wrist photoplethysmographic segments of twenty-five seconds at 32 Hz drawn from 22 subjects, each

TABLE II
CLASSIFIER TOLERANCE TO ACQUISITION TIMING ERROR

Condition	AUROC	Sens.	Spec.
Annotated beat times (reference)	0.9903	95%	97.4%
30 fps quantisation, no interpolation	0.9918	95%	96.9%
30 fps with sub-sample interpolation	0.9900	95%	96.9%
60 fps, 5 ms jitter	0.9906	95%	97.3%
15 ms jitter	0.9880	95%	95.7%
25 ms jitter	0.9826	95%	92.5%
40 ms jitter	0.9699	95%	87.4%
8 ms jitter, 5% beats omitted	0.9900	95%	96.7%

TABLE III
BEAT DETECTION AND INTERVAL ACCURACY, SMARTPHONE ACQUISITION
AGAINST SYNCHRONOUS ELECTROCARDIOGRAPHY ($n = 300$)

Subset	Sens.	PPV	RR err.	HR err.
Expert-graded good ($n=100$)	90%	90%	28.4 ms	1.0 bpm
Accepted by quality gate	85%	92%	28.9 ms	2.1 bpm
Expert-graded poor ($n=200$)	40%	57%	63.2 ms	19.7 bpm
Rejected by quality gate	44%	63%	47.6 ms	15.4 bpm

segment carrying a rhythm label and a three-level signal-quality label. The rhythm label is constant within a subject, so the effective sample size is the subject count and results are additionally aggregated per subject. The training and validation splits were not accessed.

G. Implementation Verification

The deployed implementation is JavaScript; the reference implementation is Python. Agreement is enforced by replaying 10,764 reference-computed risk values and 358 reference-computed rhythm features through the deployed code, with a tolerance of 10^{-12} . Observed worst-case divergence was 1.1×10^{-16} . Model parameters were fixed and cryptographically hashed prior to acquisition-path evaluation, and verified unchanged afterwards.

VIII. RESULTS

A. Rhythm Classification

Under leave-one-record-out validation the classifier achieved an area under the receiver operating characteristic curve of 0.9903. At an operating point selected for 95% sensitivity, specificity was 97.4%. Per-record discrimination ranged from 0.931 to 1.000.

Table II shows discrimination is preserved to approximately 25 ms of per-beat timing error, and that frame-rate quantisation alone is not limiting. Random beat omission at 5% has negligible effect, indicating that the interval-statistic descriptors are robust to sparse detection failure.

B. Acquisition Path

Table III reports performance stratified by the database's own expert quality annotations and by the system's quality gate. On expert-graded good recordings, beat-detection sensitivity and positive predictive value both reach 90%, with interval error of 28.4 ms and heart-rate error of 1.0 bpm; 83%

TABLE IV

FROZEN CLASSIFIER ON FINGERTIP PHOTOPLETHYSMOGRAPHY AT THE DEPLOYED SIXTY-SECOND WINDOW ($n = 659$ WINDOWS, 35 SUBJECTS)

Sampling rate	Sensitivity	Specificity	AUROC
30 Hz (deployed frame rate)	97.0%	95.6%	0.9918
125 Hz (native)	96.7%	94.9%	0.9902
<i>MIT-BIH, ECG intervals</i>	95%	97.4%	0.9903

TABLE V

FROZEN CLASSIFIER ON WRIST PHOTOPLETHYSMOGRAPHY, STRATIFIED BY THE CORPUS SIGNAL-QUALITY LABEL ($n = 17,614$ SCORED)

Signal quality	Segments	Sensitivity	Specificity
Excellent	3,246	91.5%	92.4%
Acceptable	2,032	91.7%	81.0%
Poor	12,336	76.1%	40.6%

of accepted recordings fall within the 5 bpm tolerance of IEC 60601-2-27.

The quality gate discriminates concordantly with the expert annotations, accepting recordings with more than twice the beat-detection sensitivity of those it rejects (85% versus 44%) and admitting expert-graded good recordings at approximately twice the rate of expert-graded poor recordings.

C. Deployed Window Length

At the deployed sixty-second configuration, specificity on continuous recordings from healthy volunteers was 91.7%—22 of 24 windows called regular. This is a direct measurement at the operating configuration rather than an extrapolation from shorter clips, and it agrees to within 0.7 percentage points with the 92.4% projected below from acquisition error measured on an unrelated corpus.

D. Rhythm Classification at the Deployed Configuration

Table IV evaluates the frozen classifier at both the deployed window length and the deployed anatomical site. Sensitivity is 97.0% (95% CI 94.7–98.3%) and specificity 95.6% (92.6–97.4%) at 30 Hz, with an area under the curve of 0.9918 against 0.9903 on electrocardiographic intervals. Per subject, all 19 fibrillatory subjects and 15 of 16 sinus-rhythm subjects were classified correctly. Every window was scored; none was rejected as unusable.

Decimating from 125 Hz to 30 Hz changes sensitivity by 0.3 points and specificity by 0.7—within the confidence intervals of both. Camera frame rate is therefore not the limiting factor in interval-based rhythm discrimination, which is the outcome the sub-sample interpolation of Section V-B is intended to produce, measured here rather than assumed.

E. Rhythm Classification on Wrist Photoplethysmographic Intervals

Table V reports the frozen classifier applied to intervals derived from a pulse wave rather than from electrocardiography. On excellent-quality segments sensitivity is 91.5% and

TABLE VI

PROJECTED PREDICTIVE VALUES BY SCREENING PREVALENCE

Population (AF prevalence)	PPV	NPV
General adults, age ≥ 55 (2%)	20.4%	99.89%
Adults, age ≥ 65 (5%)	39.8%	99.72%
Higher-risk, age ≥ 75 (10%)	58.3%	99.40%

specificity 92.4%, against 95% and 97.4% on electrocardiographic intervals: the transition from QRS timing to pulse-wave timing, with its softer peaks and respiratory amplitude modulation, costs approximately three points of sensitivity and five of specificity. Performance declines monotonically with signal quality, and 70% of this corpus is graded poor, reflecting continuous wrist wear rather than deliberate fingertip acquisition.

Aggregated per subject over excellent-quality segments, all four fibrillatory subjects and all six sinus-rhythm subjects were classified correctly. With ten subjects these are reported as counts rather than as rates.

F. Projected Screening Performance

The measured interval error of 28.4 ms corresponds, under a Gaussian error model, to a per-beat timing dispersion of approximately 25 ms. Interpolating Table II at that operating point projects a specificity of 92.4% at 95% sensitivity.

Table VI gives the resulting predictive values. These fall within the 20–40% positive and near-100% negative predictive values reported for validated smartphone screening applications [5], despite being derived independently—from classifier performance on electrocardiographic data combined with acquisition error measured on a separate smartphone corpus. The agreement between two unrelated derivations supports the validity of the projection.

The photoplethysmographic operating point of Table V provides a third, independent route to the same quantity: 91.5% sensitivity at 92.4% specificity yields 19.7% positive and 99.81% negative predictive value at 2% prevalence, within one percentage point of Table VI despite deriving from a different corpus, a different anatomical site and a different sensing modality.

Table VI is nonetheless conservative. Applying the measured fingertip operating point of Table IV in place of the projection gives 31.0% positive and 99.94% negative predictive value at 2% prevalence. The qualitative asymmetry that governs the decision rule is unchanged: a negative result remains dependable and a positive result remains, more often than not, wrong.

These values establish the decision rule of Section V-E quantitatively. A negative predictive value of 99.9% renders exclusion of atrial fibrillation dependable; a positive predictive value of 20% at population prevalence renders assertion unsupportable. The module is therefore permitted to exclude and required to refer.

G. Risk Model Behaviour

Reconstruction reproduces the published five-year cohort risks to within 0.002 for both sexes and both coefficient sets. Risk is monotone in age, systolic pressure and each binary factor. For a 68-year-old man with no other information supplied, the interval estimate spans 2.2–5.0%, and the variance decomposition attributes 52% of that uncertainty to smoking status and 10% to systolic pressure; for a 58-year-old woman the ranking inverts, with systolic pressure accounting for 36%, reflecting the steeper pressure gradient in the female coefficient set. The recommendation is thus subject-specific rather than a fixed ordering.

IX. DISCUSSION

The central design question for a self-administered risk instrument is not which additional sensor to add, but how to behave when required inputs are unavailable. Substituting population means for unknown inputs produces a confident number from fabricated data. Representing them as distributions produces a wider but honest estimate, and—through variance decomposition—an actionable statement of which single measurement would most reduce that width. This reframes missing data from a defect of the instrument into its primary output.

The quality-gating result of Section V-C generalises beyond this system. Quality metrics are commonly transferred between photoplethysmographic tasks on the assumption that a clean signal is clean for all purposes. Where the target phenomenon is itself a departure from regularity, metrics defined in terms of regularity—spectral peak sharpness, rate stability—cannot separate signal from artefact. Any pipeline gating arrhythmia analysis with heart-rate quality criteria will preferentially discard its positive cases, and will do so silently.

The predictive-value asymmetry has a design consequence frequently left implicit. A screening instrument reporting 95% sensitivity and 92% specificity is legitimately described as accurate, yet at 2% prevalence four of five positive indications are false. Systems that report such a result as a finding will mislead the majority of those they flag. Reporting it as a referral preserves the clinical value of the true positives while making no claim the evidence cannot support.

Finally, population-specific extension is achievable without abandoning a validated equation. Adding smokeless tobacco as a zero-centred adjunct term with a published adjusted hazard ratio leaves the derivation-cohort calibration intact while capturing an exposure affecting 29.6% of men and 12.8% of women in the target population [13], which the underlying Western-derived model does not represent at all.

X. LIMITATIONS AND FUTURE WORK

Sensitivity and specificity derive from distinct corpora. Classifier sensitivity is established on electrocardiographic intervals and corroborated on both fingertip and wrist photoplethysmographic intervals; acquisition fidelity and specificity are established on smartphone fingertip recordings. No public corpus provides smartphone *fingertip* photoplethysmography

with adjudicated fibrillation labels, so the end-to-end performance of Table VI remains assembled from separate measurements rather than observed directly. A prospective acquisition with synchronous electrocardiography would resolve this and is the principal item of future work.

Dependence on signal quality. Table V shows specificity falling to 40.6% on segments the corpus grades poor. Reported performance is therefore conditional on the quality gate admitting only adequate recordings. That gate was characterised on deliberate fingertip acquisition, and its behaviour on continuous wrist wear—the modality of Table V—has not been separately quantified.

Analysis window. Classification has been measured at the deployed sixty-second window on fingertip photoplethysmography (Table IV) and on continuous smartphone recordings from healthy volunteers, but the smartphone *camera* acquisition corpus provides only ten-second clips. The step that remains inferred rather than observed is therefore the camera itself, not the window length or the sensing site.

Cohort representativeness. The acquisition corpus comprises fifty volunteers with a mean age of 33 years, and does not report Fitzpatrick skin phototype. Since melanin attenuates the optical signal, performance across skin tone remains uncharacterised and warrants explicit stratification in prospective evaluation.

Derivation range and geography. The risk equation is derived on adults aged 55–84 in United States cohorts; the system labels estimates outside that range as extrapolation. Relative effects transport between populations more reliably than absolute baselines, and recalibration of S_0 to age- and sex-specific local incidence is a data requirement rather than a modelling one. Reported calibration for this model family is weaker in Black participants than in white participants, and the system discloses this in its output.

Adjunct term independence. Adjunct hazard ratios are drawn from meta-analyses with heterogeneous covariate adjustment, and are combined multiplicatively with the base model under an independence assumption that is defensible where the exposure is absent from the derivation cohort, but is not directly verified.

Regulatory scope. Presentation of an absolute stroke probability to a lay user constitutes a medical device claim in most jurisdictions. The system is presented as a research instrument and is not offered for clinical use.

XI. CONCLUSION

We have presented a multimodal stroke risk system producing a calibrated five-year absolute probability from a questionnaire, a sixty-second fingertip photoplethysmogram, and optional medical document capture, executing entirely on a consumer smartphone with no data transmission. Absolute calibration is anchored to a published prospective-cohort survival model with fully traceable coefficients; unknown inputs are propagated as uncertainty and ranked by variance contribution rather than imputed; and rhythm assessment employs quality

criteria constructed specifically for arrhythmia rather than inherited from heart-rate estimation.

Rhythm classification achieves an area under the curve of 0.990 under leave-one-patient-out validation on electrocardiographic intervals, and 0.992 at the deployed sixty-second window on fingertip photoplethysmography from an independent corpus, where sensitivity and specificity are 97.0% and 95.6%. The acquisition path achieves 90% beat-detection sensitivity with 28.4 ms interval error on expert-graded smartphone recordings. That the fingertip result is unchanged by decimation to 30 Hz indicates camera frame rate is not the limiting factor in interval-based rhythm discrimination. The resulting predictive-value asymmetry—above 99.9% negative against 20–31% positive at population prevalence—directly determines a decision rule under which the system may exclude atrial fibrillation from the risk computation but refers rather than asserts it. We argue this pattern, in which the evidentiary strength of a screening result governs the action it is permitted to take, is the appropriate organising principle for consumer-facing diagnostic inference.

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